

FondoMexicoAlfa (FMIA)

Systematic Long-Only Equity & FIBRA | Quantitative Research Report

0.03	8.46%	13.70%	-37.19%	-1.98%	8.87%
Sharpe Ratio	Annualized Return	Annualized Vol	Max Drawdown	CVaR 95% (daily)	Avg Turnover/Rebalance

Headline metrics from live run: source=bloomberg, model=elasticnet, as_of=2026-03-31.

Walk-forward validated * Bloomberg data * Mexican equities & FIBRAs (BMV) * Monthly rebalancing * 10 bp transaction cost

Report date: 2026-05-19

Strategy

Universe:	~26 Mexican equities + FIBRAs listed on BMV
Signal:	LightGBM cross-sectional return forecast (RandomizedSear
Baseline:	ElasticNetCV - interchangeable via config flag
Optimizer:	Mean-variance (SLSQP) with market-impact penalty
Rebalance:	Monthly, end-of-month, walk-forward OOS
Validation:	108 rebalances, 2017-2026

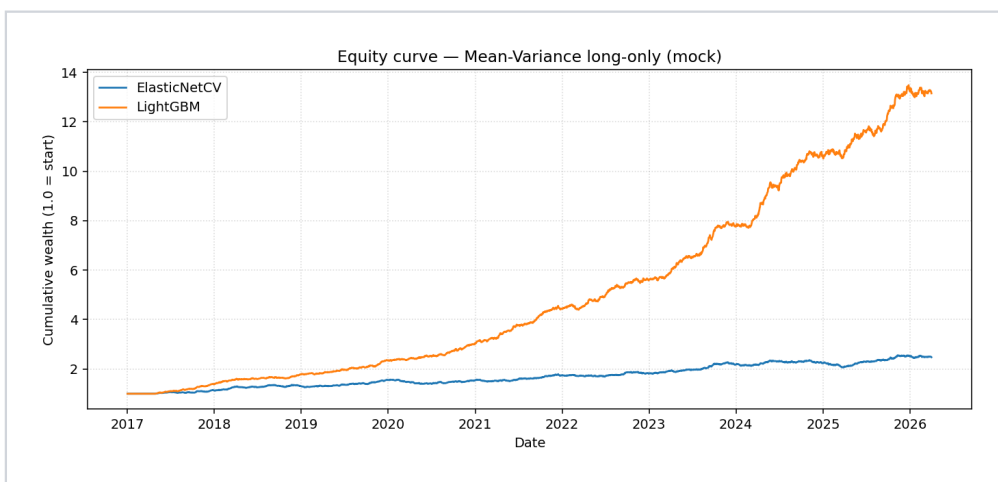
Risk Controls & Attribution

Costs:	10 bp per side applied each rebalance
Regimes:	Banxico rate (TIGHTENING/EASING/NEUTRAL) x IPC vol stres
Attribution:	Per-rebalance SHAP values via TreeExplainer
Safety valve:	Revert to ElasticNet if top-5 SHAP stability < 0.30 for
Data:	Bloomberg (production) / mock data (research)
Framework:	CNBV position limits, LFI structure

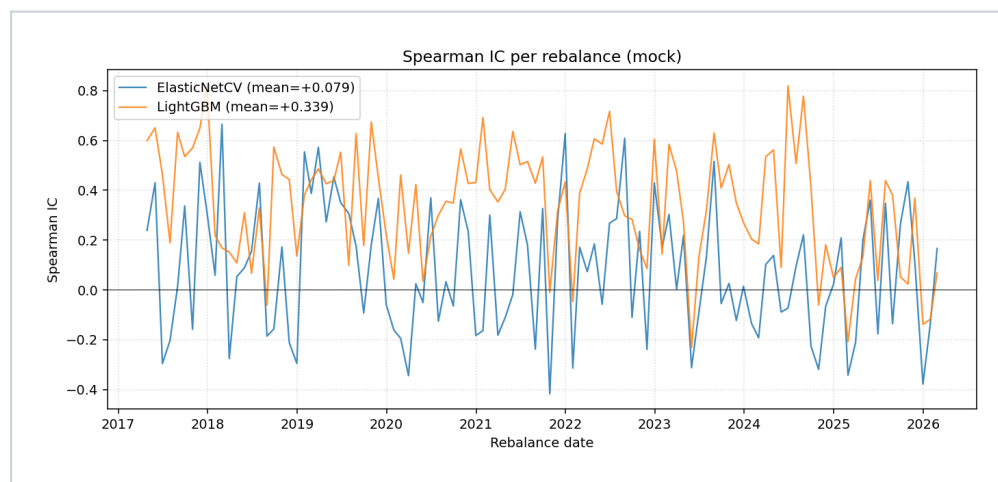
Headline metrics from live run: source=bloomberg, model=elasticnet, as_of=2026-03-31.

01 MODEL PERFORMANCE - LightGBM vs ElasticNetCV (mock data - pipeline validation only)

Metric	ElasticNetCV	LightGBM	D (lgbm - elastic)
IC mean (Spearman)	+0.0794	+0.3389	+0.2595
IC std	0.2597	0.2335	-0.0263
ICIR	+0.306	+1.452	+1.1460
Hit rate (daily)	0.516	0.573	+0.0564
Annualized return	+0.0995	+0.3089	+0.2094
Annualized vol	0.0763	0.0757	-0.0006
Sharpe	+0.981	+3.292	+2.3100
Sortino	+1.025	+3.626	+2.6012
Max drawdown	-0.127	-0.046	+0.0806
CVaR 95% (daily)	-0.0097	-0.0091	+0.0006
Turnover (per rebalance)	0.0624	0.2684	+0.2060
Forecast wall time (s)	13.0	573.0	+560.0181



Cumulative equity curves (108 rebalances)



Rolling information coefficient (IC) time series

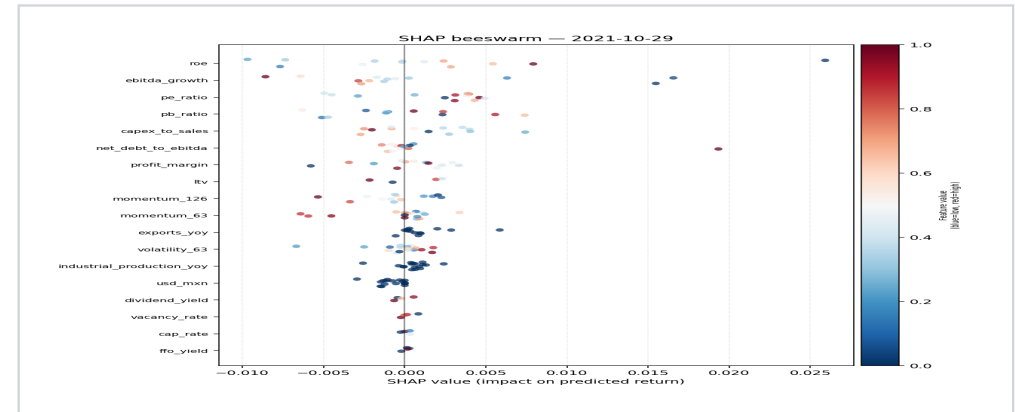
02 FEATURE ATTRIBUTION (SHAP) - LightGBM Walk-forward

Top-10 Features by Mean |SHAP|

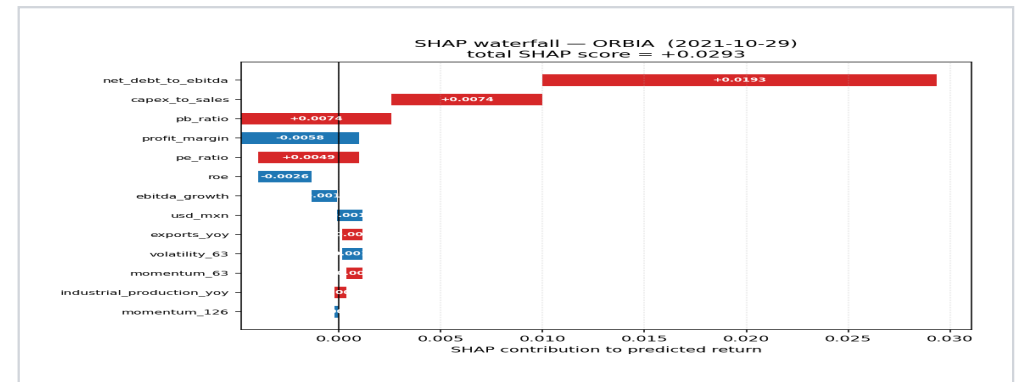
#	Feature	Mean SHAP	Std
1	ltv	0.00350	0.00363
2	ffo_yield	0.00279	0.00360
3	cap_rate	0.00221	0.00235
4	pe_ratio	0.00196	0.00233
5	dividend_yield	0.00182	0.00214
6	momentum_63	0.00172	0.00249
7	roe	0.00154	0.00183
8	ebitda_growth	0.00146	0.00188
9	profit_margin	0.00145	0.00220
10	capex_to_sales	0.00136	0.00219

Feature Stability (Spearman rank-correlation)

K	Pairs	Mean r	Std r
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top-5	107	0.440	0.421
top-10	107	0.428	0.329
all	107	0.455	0.292



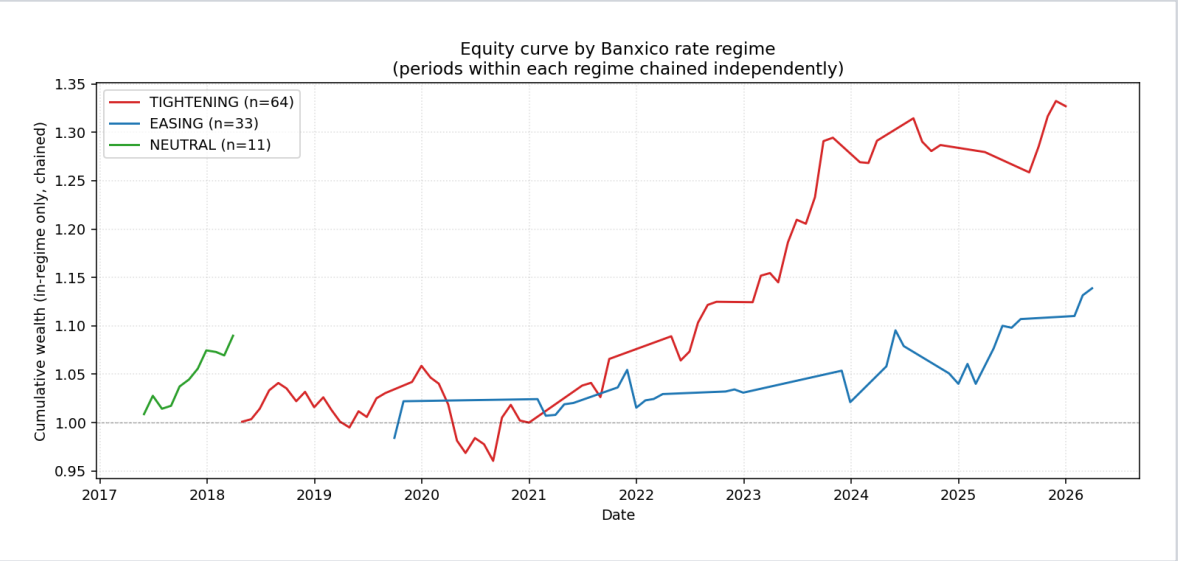
Beeswarm - Midpoint Rebalance (2021-10-29)



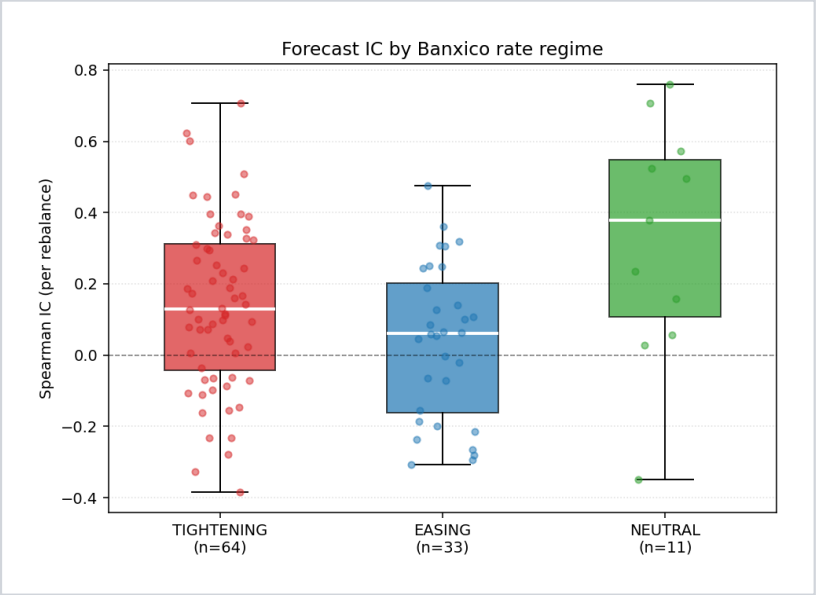
Waterfall - Highest-Predicted Stock

03 PERFORMANCE BY MACRO REGIME (mock data - pipeline validation only)

Rate regime	Stress	N	IC mean	Sharpe	SHAP top-5 stab.
EASING	ALL	33	+0.039	+1.27	0.566
EASING	CALM	23	+0.021	+1.74	-
EASING	STRESS	10	+0.115	+1.58	-
NEUTRAL	ALL	11	+0.280	+2.27	0.386
NEUTRAL	CALM	11	+0.280	+2.27	-
TIGHTENING	ALL	64	+0.140	+1.01	0.407
TIGHTENING	CALM	47	+0.195	+1.07	-
TIGHTENING	STRESS	17	-0.035	+1.66	-



Regime-conditioned cumulative equity curves



IC distribution by macro regime

04 RISK PROFILE & METHODOLOGY

ElasticNet Regulated - Bloomberg OOS (source of truth)

Metric	Value
Annualized return	8.46%
Annualized vol	13.70%
Sharpe ratio	0.03
Max drawdown	-37.19%
CVaR 95% (daily)	-1.98%
Avg turnover/rebalance	8.87%

Headline metrics from live run: source=bloomberg, model=elasticnet, as_of=2026-03-31.

LightGBM Risk Metrics (Walk-forward OOS - mock data)

Metric	Value
Annualized return	+0.3089
Annualized vol	0.0757
Sharpe ratio	+3.292
Sortino ratio	+3.626
Max drawdown	-0.046
CVaR 95% (daily)	-0.0091
Avg turnover/rebalance	0.2684

Methodology

Universe:	Mexican equities + FIBRAs (BMV), ~26 instruments
Signal:	LightGBM cross-sectional return forecast (RandomizedSearchCV + TimeS
Baseline signal:	ElasticNetCV (interchangeable via config flag)
Optimizer:	Mean-variance (SLSQP) with market-impact penalty
Rebalance:	Monthly (end-of-month), walk-forward OOS
Validation:	108 rebalances, 2017-2026
Costs:	10 bp per side applied each rebalance
Regimes:	Banxico rate (TIGHTENING/EASING/NEUTRAL) x IPC vol stress
Attribution:	Per-rebalance SHAP via TreeExplainer (shap 0.51)
Data:	Bloomberg (production) * mock data (research)
Framework:	CNBV position limits, LFI structure

DISCLAIMER: For research purposes only. Past performance does not guarantee future results. All results based on walk-forward out-of-sample simulation. Mock data used where Bloomberg data unavailable. This document is confidential and intended solely for internal research use.